Course Outline for CSE-331

Part A

1. Course Code: CSE-331

2. Course Title: Theory of Computing

3. Course Type: Core Course

4.Level/ Term: Level: 3 Term: I

5. Academic Session: 2020-21

6. Course Teacher: Ashim Dey, Assistant Professor, Dept. of CSE, CUET

7. Prerequisite(s): None

8. Credits: 2

9. Contact Hours: 2 (2 lectures of 50 minutes per week)

10. Total Marks: 200

11. Rational of the Course:

This course will cover the advanced knowledge of formal computation and its relationship to languages and basis of theory of computation, in particular the role of key problems in defining classes of equivalent problems from a computational perspective. This course is important to understand the role of computation theory and the limitations of computational procedures. This a required course for all the students enrolling B. Sc. Engg. in CSE program. The catalogue description of the course is

Course Content:

Central concept of Automata Theory: Alphabets, Strings, Length of Strings, Power of Alphabets, Languages, Problems, Define Language using Set Former; Finite Automata: Basic Concepts of DFA & NFA, Designing DFA & NFA, Transition Diagram, Transition table, NFA to DFA conversion, Computation of ε-closure of states, ε-NFA, Equivalance of ε-NFA to NFA. Regular Expressions: Basic concepts, Application of regular expressions, Conversion of Regular expressions to NFA, Conversion of NFA/DFA to Regular expressions, Context Free Grammars (CFGs): Definition/Properties of CFG, Designing VFG, Left-most Derivation, Right-most Derivation, Parse tree, Equivalence of derivation & Parse tree, Ambiguity and Ambiguous CFGs, Conversion og Ambiguous CFGs to

Unambiguous CFgs. Push Down Automata (PDA): Basic Concepts, Transition Function, Transition Diagram, Designing PDA, Instantaneous Description of PDA, Languages of PDA and Their Conversion, Conversion of PDA to CFG & CFG to PDA, Chomsky Normal Form (CNF): Basic properties and conversion of CFG to CNF, Pumping Lemma, Turing Machines (TM): Basic concepts, Configuration, Transition Function, Transition Diagram, Instantaneous Description of TM, Computing and designing Turing Machines, Halting problem of TM, Programming techniques for TM (Multi-track TM, Multi-tape TM, Conter Machines), Simulating TM, Combining TM, Comparing running times of computers and TM, Undecidability: Basic concepts, Decidable Languages, Reducibility, Codes of TM, Hilbert's problems.

12. Course Objectives:

(a) Demonstrate advanced knowledge of formal computation and its relationship to languages.

(b) Analyze and design finite automata, pushdown automata, turing machines, formal languages and grammars.

(c) Demonstrate their understanding of key notions, such as algorithm, computability, decidability and complexity through problem solving.

(d) Understand the basis of theory of computation, in particular the role of key problems in defining classes of equivalent problems from a computational perspective.

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

	Course Learning Outcomes (CLOs)	**Blooms Level (Optional)**
CLO1	Explain the models of computation, including formal languages, grammars and automata, and their connections.	C2,C5,C6,A3,A4,P2
CLO2	Analyze finite automata, pushdown automata, Turing machines, formal languages and grammars with specific properties.	C4
CLO3	Solve computational problems regarding their computability and complexity and prove the basic results of the theory of computation.	C3,A5

b) Mapping of CLO with PLO

No.	(CLOs)	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
1	CLO1	X											
2	CLO2		X										
3	CLO3		X										

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

`	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Week-01	Overview. outlines & Importance of theory of computation ; Basic Mathematical Notation and techniques, Finite State systems - Basic Definitions	• Lecture	Class Test Final Exam	CLO-1
Week - 02	Finite Automaton –DFA & NFA difference between NFA &DFA, Equivalence of NFA to DFA, Equivalence DFA to NFA	• Lecture	Class Test Final Exam	CLO-2

Week - 03	Regular Languages- Regular Expression (RE), properties of RE.	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-1
Week - 04	Finite Automata & Regular Expressions, Closure properties of regular Languages.	• Lecture	Class Test Final Exam	CLO-2
Week - 05	Equivalence of Automata and regular expressions	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-2
Week - 06	Context Free Grammars (CFGs) and Context Free Languages (CFLs): Basic properties, importance, applications.	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-2
Week - 07	Derivations and Languages, Top-down vs. and bottom-up parsing, parse tree. Ambiguity, ambiguous grammar, Relationship between derivation and derivation parse trees.	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-2

Week - 08	Simplification of CFG; Conversion of CFG to CNF and GNF.	• Lecture	Class Test Final Exam	CLO-3
Week - 09	Elimination of useless symbols, Unit productions, Null productions: Theory and Illustrative examples.	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-3
Week - 10	Pushdown Automata (PDA)- Definition, Graphical notation PDA design, Instantaneous Descriptions (IDs) and problems	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-3
Week - 11	Introduction- formal definition of Turing machines (TM),	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-3
Week - 12	Undecidability, Basic definitions Decidable and undecidable problems	• Lecture	Class Test Final Exam	CLO-1
Week - 13	Introduction to computational time & space complexity - complexity, Definitions:of TMs, Introduction	• Lecture	Class Test Final Exam	CLO-3

	to NP-hardness & NP Completeness			

Part C

15. Assessment and Evaluation

l) Assessment Strategy

Class participation and attendance	10%
Class tests/Class assessment	20%
Term Final Examination (3 hours duration)	70%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 30%

b) Summative: 70%

c) Make-up Procedures:

- Feedback on continuous assessment is given to the students immediately after the test.
- The minimum number of class-test/assignment are (n+1) with best n will be counted (here, n is number of credit). Based on the students' feedback additional class-test/assignment may be taken by the course teacher

Part D

16. Learning Materials

1) Recommended Readings

- E.Hopcroft and J.D.Ullman, "Introduction to Automata Theory, Languages and Computation", Pearson, Education Publishers, 2nd Edition, 2004.

2) Supplementary Readings

- Michael Sipser, "Introduction to the Theory of Computation". Thomson Asia, 2004.

2. J.C.Martin,

- "Introduction to languages and Theory of Computation", McGraw Hill, 2003.

3. K.L.P. Mishra, and N.Chandrasekaran,

- “Theory of Computer Science: Automata, Languages and Computation", 3rd edition, PHI Learning Pvt. Ltd. 2006.

3) Others

- Handout/lecture material provided by the course teacher